DIGITAL ELECTRONICS LAB (ESC392)

LAB ASSIGNMENT 6 — APPLICATION MAPPING OF SEQUENTIAL BUILDING BLOCKS

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1. Aim / Objective

To identify real-world applications of sequential building blocks—flip-flops, Mod-N counters, and shift registers—and explain their roles in practical digital systems.

2. Software and Tools Required

1. Logisim Evolution
2. Digital logic components: flip-flops, logic gates, counters, and shift registers
3. Microsoft Word or Google Docs for report preparation

3. Application 1: Flip-Flop as a 1-Bit Memory Cell

Real-World System

CPU register or digital data-storage circuit

Description

A flip-flop is a sequential circuit that stores one bit of information, either 0 or 1. Its output remains stored until a suitable clock edge or control signal changes its state.

In a CPU, multiple flip-flops can be connected to form a register that temporarily stores data during processing. Flip-flops are also used in control circuits and digital systems that need to remember a previous state.

Working Principle

- Apply the data bit to the D input.
- Apply a clock pulse.
- On the active clock edge, the flip-flop stores the input value.
- The Q output holds that value until it is updated.

Simulation in Logisim Evolution

1. Place one D flip-flop.

2. Connect an input pin to D.
3. Connect a clock to the clock input.
4. Connect an output pin or LED to Q.
5. Change the input and apply a clock pulse.
6. Observe that Q stores the input value after the active clock edge.

Justification

The D flip-flop represents a one-bit memory element. Connecting several flip-flops allows a digital system to store multiple bits of data.

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4. Application 2: Mod-60 Counter

Real-World System

Seconds counter in a digital clock

Description

A Mod-60 counter counts 60 states, from 0 to 59, and then returns to 0. It is used in digital clocks to count seconds or minutes. It can also be used as a frequency divider or timing element in a control system.

Working Principle

- The counter begins at 0.
- Each clock pulse increases the count by 1.
- When the count reaches 59, the next clock pulse returns it to 0.
- The counting sequence repeats continuously.

A 6-bit binary counter is sufficient to represent the numbers 0–59 because six bits can represent values from 0 to 63.

Simulation in Logisim Evolution

1. Place six D flip-flops to form a 6-bit counter.
2. Connect all flip-flops to a common clock.
3. Build the required counter logic using XOR and AND gates.
4. Detect the binary value of decimal 59, which is 111011 when written as Q5 Q4 Q3 Q2 Q1 Q0.

5. Add suitable synchronous reset logic so that the next state after 59 is 0.
6. Observe the sequence and verify that it repeats from 0 to 59.

Justification

The Mod-60 counter provides the repeating 60-state sequence needed to measure seconds or minutes in a digital clock. The same principle can be adapted for other timing applications.

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5. Application 3: Shift Register

Real-World System

Serial-to-parallel data conversion in digital communication

Description

A shift register is a sequential circuit consisting of flip-flops connected in a chain. It shifts stored data from one stage to the next whenever a clock pulse is applied.

A Serial-In Parallel-Out (SIPO) shift register receives data one bit at a time and makes the collected bits available simultaneously at its parallel outputs. A Parallel-In Serial-Out (PISO) shift register performs the reverse operation.

Working Principle of a SIPO Register

- Serial data enters the first flip-flop.
- Each clock pulse shifts the stored bits to the next stage.
- After the required number of clock pulses, the stored bits appear at the parallel outputs.

Simulation in Logisim Evolution

1. Place four D flip-flops in a row.
2. Connect the Q output of each stage to the D input of the next stage.
3. Connect all clock inputs to the same clock.
4. Connect a serial input to the D input of the first flip-flop.
5. Connect output pins or LEDs to the Q outputs.
6. Apply serial data bits and clock pulses.
7. Observe how the data shifts through the four stages.

Justification

A shift register converts serial data into parallel data, which is useful in digital communication and data-processing circuits. It can also be used in display-driving and data-transfer applications.

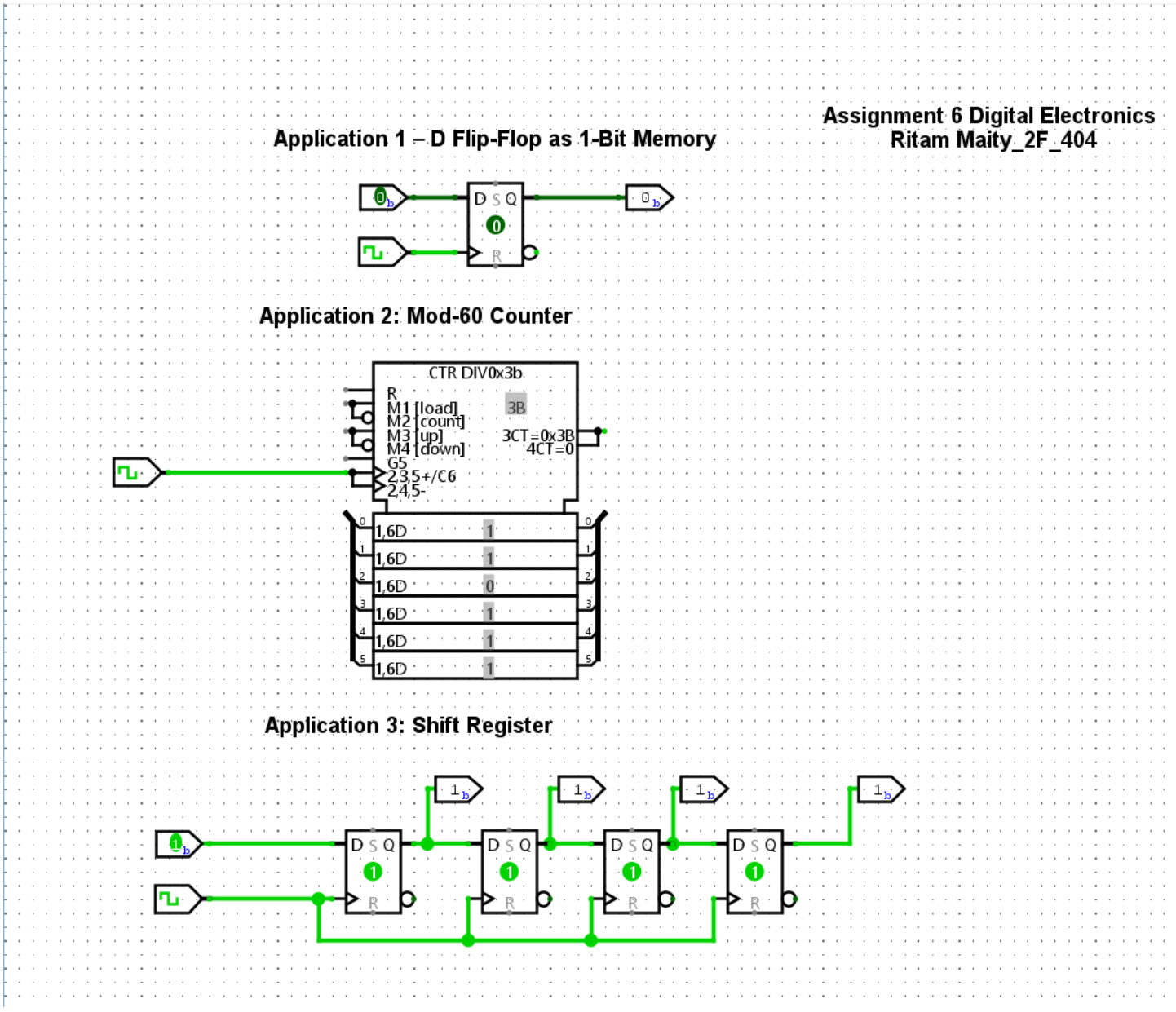
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6. Comparative Summary

Sequential block	Main function	Real-world application
D flip-flop	Stores one bit	CPU register
Mod-60 counter	Counts 0–59 repeatedly	Digital clock
Shift register	Stores and shifts data	Serial-to-parallel conversion

7. Result

The real-world applications of three sequential building blocks were identified and explained. A flip-flop stores one bit, a Mod-60 counter provides a repeating 60-state sequence, and a shift register transfers data between serial and parallel formats.



8. Discussion

Sequential circuits use memory elements and clock signals to store data or perform operations over time. Their applications include data storage, digital timing, and communication systems. Simulation helps verify their behaviour before implementation in practical hardware.

9. Conclusion

This assignment demonstrates the importance of sequential building blocks in real digital systems. Flip-flops, counters, and shift registers perform different but essential functions in storage, timing, and data transfer.

10. GitHub Repository Link

GitHub URL: https://github.com/Ritam-Maity/Digital-Electronics-Lab